[University name]

Air Pollution and Respiratory Morbidity in Kampala, Uganda: An Ecological Time-Series Analysis of Weekly Routine Surveillance Data

A thesis chapter submitted in partial fulfilment of the requirements for the degree of *[PhD degree]*

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[Month Year]

Abstract

Background. Fine particulate air pollution is an important environmental health concern, but evidence based on routinely collected morbidity data in East African cities remains limited. This chapter examined weekly associations between ambient $PM_{2.5}$ concentrations and four respiratory outcomes in Kampala, Uganda.
Methods. We linked weekly climate, averaged $PM_{2.5}$, and routine respiratory surveillance records from 5 January 2020 to 15 December 2024 (248 weeks). Outcomes were tuberculosis (TB), influenza-like illness (ILI), severe pneumonia in children younger than five years, and severe acute respiratory infection (SARI). Robust-standard-error Poisson models estimated adjusted incidence rate ratios (IRRs) per 10 μg/m^3 higher $PM_{2.5}$, controlling for temperature, relative humidity, precipitation, calendar month, and linear time. Exploratory 0–4-week distributed-lag, climate, sensitivity, seasonal, and spatial analyses complemented the primary models.
Results. Mean weekly $PM_{2.5}$ was 39.28 μg/m^3 (SD 11.81). Higher $PM_{2.5}$ was associated with higher ILI notifications (IRR 1.173, 95% CI 1.030–1.335; $p = 0.016$), but not with SARI (1.038, 0.950–1.135) or severe pneumonia in children under five (0.956, 0.908–1.005). TB showed an inverse association (0.952, 0.924–0.981), which should not be interpreted as protective. The cumulative 0–4-week association was positive for ILI (1.192, 1.032–1.377) and compatible with the null for other outcomes. Results for ILI remained similar after excluding the highest 5% of $PM_{2.5}$ observations and after using a one-week lag.
Conclusions. Routine data revealed an outcome-specific association between $PM_{2.5}$ and ILI, but the ecological design, incomplete control of confounding, exposure aggregation, and surveillance artifacts preclude causal attribution. The findings support improved integrated environmental and health surveillance and prospective, pre-specified studies.

Keywords: particulate matter; air pollution; respiratory infection; routine surveillance; time series; Kampala; Uganda; ecological epidemiology.

Abbreviations

$PM_{2.5}$	Particulate matter with aerodynamic diameter $\leq 2.5\,\mu$m
CI	Confidence interval
ILI	Influenza-like illness
IRR	Incidence rate ratio
SARI	Severe acute respiratory infection
TB	Tuberculosis

Contents

Chapter 1

Introduction

1.1 Background and rationale

Ambient particulate matter is a policy-relevant exposure because inhalable particles can reach the lower respiratory tract and are linked to a wide range of adverse health outcomes. The World Health Organization (WHO) provides health-based guideline levels and interim targets for $PM_{2.5}$ and other pollutants (**WHO2021**). Kampala faces traffic, road dust, combustion, and urban-development pressures that make local, decision-relevant evidence important. Yet a city-wide association estimated from routine data must be interpreted cautiously: notification counts reflect disease occurrence, care seeking, diagnostics, reporting, and health-service disruptions as well as environmental conditions.

This chapter uses project-supplied weekly records to examine whether variation in ambient $PM_{2.5}$ coincided with variation in four respiratory morbidity outcomes. It follows observational-reporting principles in the STROBE statement (**vonElm2007**) and separates association estimates from causal and policy claims.

1.2 Research questions and objectives

The primary research question was: *How were weekly ambient $PM_{2.5}$ concentrations associated with weekly notifications of TB, ILI, severe pneumonia in children under five, and SARI in Kampala?*

The objectives were to: (i) describe the time series and seasonal patterns; (ii) estimate adjusted contemporaneous associations; (iii) examine 0–4-week lagged associations; (iv) test the stability of estimates in prespecified sensitivity analyses; and (v) describe geographic variation in available monitoring and health-record data.

1.3 Conceptual framework

Emission sources and urban activity (traffic, dust, combustion) $\rightarrow$ **ambient $\mathbf{PM_{2.5}}$** $\rightarrow$ **respiratory susceptibility, symptoms, and illness** $\rightarrow$ **health-care attendance, diagnosis, and notification**.

Temperature, humidity, precipitation, seasonality, secular trends, health-service access, diagnostic practice, and reporting completeness may affect both exposure patterns and notified morbidity.

Figure 1.1: Analytic conceptual framework. The routinely collected outcome is a notification count, not a direct measure of disease incidence.

Chapter 2

Methods

2.1 Design, setting, and analytic period

This was an ecological weekly time-series analysis in Kampala, Uganda. The analytic period covered 5 January 2020 through 15 December 2024 and contained 248 reporting weeks. The unit of analysis was the calendar week; therefore, individual-level causation cannot be inferred from the study estimates.

2.2 Data sources and linkage

Three project-supplied sources were linked by epidemiological year and week or by the weekly ending date: (i) respiratory surveillance records, (ii) climate records containing temperature, humidity, wind speed, and precipitation, and (iii) weekly averaged $PM_{2.5}$ observations. The analytic dataset was constructed in `analysis_no_deaths.ipynb`; its provenance and source files are documented as project data (**ProjectData2026**). The spatial description additionally used two supplied monitor-export files, a Kampala administrative-boundary shapefile, and a supplied facility/patient abstraction workbook.

All mortality fields were excluded before analysis. The four retained outcomes were: Pulmonary TB cases (TB), ILI cases, severe pneumonia under five cases, and SARI cases. Counts were modeled as supplied; no population denominators or offsets were available, so results describe associations with notified counts rather than incidence rates in the population.

2.3 Data processing and feature construction

Weekly dates were converted to dates, and duplicate climate/air-quality keys were reconciled during the merge. Mean imputation was applied to $PM_{2.5}$, temperature, humidity, and the included outcome columns when missing; the final model-specific records had no missing values for their included terms. The resulting series had 11 missing calendar weeks according to the processor log, so the time scale represents observed reporting weeks rather than an uninterrupted daily exposure series.

For exploratory analyses, the notebook generated 1–4-week $PM_{2.5}$ lags, rolling averages, outcome changes, temperature–humidity interactions, and rolling weather summaries. These features were used for descriptive and exploratory analyses; they were not all entered simultaneously in the primary regression.

2.4 Primary statistical model

For outcome Y_t at week t, the primary Poisson generalized linear model was

$$\log\{E(Y_t)\} = \beta_0 + \beta_1 \mathrm{PM}_{2.5,t} + \beta_2 T_t + \beta_3 H_t + \beta_4 P_t + \beta_5 \mathrm{month}_t + \beta_6 t.$$

The model included ambient $\mathrm{PM}_{2.5}$, temperature (T), relative humidity (H), precipitation (P), calendar month, and linear time. Robust HC3 standard errors were used. Estimates are expressed as $\exp(10\beta_1)$, the IRR associated with a 10 μg/m^3 higher weekly $\mathrm{PM}_{2.5}$ concentration. Poisson regression is a standard framework for count outcomes, while valid inference depends on appropriate mean specification and variance handling (**CameronTrivedi2013**).

2.5 Secondary and sensitivity analyses

An exploratory distributed-lag model entered separate linear $\mathrm{PM}_{2.5}$ terms at lags 0–4 weeks, adjusted for the same weather, month, and time terms. Lag-specific coefficients were summed with their covariance matrix to calculate a cumulative 0–4-week IRR. This simplified linear-lag specification is related to, but not a full implementation of, a distributed lag non-linear model (**Gasparrini2010**).

Climate models reported temperature per 1°C and relative humidity per 10 percentage-point increment. Sensitivity analyses (a) excluded the highest 5% of $\mathrm{PM}_{2.5}$ weeks and (b) replaced contemporaneous $\mathrm{PM}_{2.5}$ with a one-week lag. Monthly and seasonal summaries were descriptive. Spatial analyses mapped division-level mean monitored $\mathrm{PM}_{2.5}$ and health-record counts; because the sources are not aligned individual-level panels, these maps are descriptive only.

2.6 Exploratory predictive and intervention analyses

The notebook also fitted LSTM-based models with an eight-week sequence, a nominal 80/20 temporal split, and environmental/time features. It benchmarked classical classifiers separately. A midpoint date of 29 May 2022 was mechanically designated as an intervention reference, not an observed policy intervention. Scenario calculations applied a fixed assumed relative risk of 1.08 per 10 μg/m^3 change to hypothesized $\mathrm{PM}_{2.5}$ reductions. These analyses are presented in Appendix B; they were not used to support the principal epidemiological conclusions.

Chapter 3

Results

3.1 Analytic dataset

The analytic series included 248 weeks. Mean $PM_{2.5}$ was 39.28 $\mu g/m^3$ (SD 11.81; range 16.87–90.92). Mean weekly notified counts were 638.26 for TB, 339.95 for ILI, 124.11 for severe pneumonia in children under five, and 220.12 for SARI (Table 3.1). ILI showed substantially greater relative variability than the other outcomes, consistent with episodic reporting or disease peaks.

Table 3.1: Descriptive statistics for weekly analytic variables, Kampala, 2020–2024.

Variable	Mean	SD	Minimum	Median	Maximum
$PM_{2.5}$ ($\mu g/m^3$)	39.28	11.81	16.87	38.06	90.92
Temperature (°C)	22.85	0.97	20.83	22.72	27.04
Relative humidity (%)	80.96	5.05	54.71	81.60	90.06
Precipitation (mm/day)	28.21	27.76	0.02	19.16	133.74
TB cases	638.26	251.80	7	638.26	1321
ILI cases	339.95	718.62	1	40.50	3498
Severe pneumonia, under 5	124.11	75.61	4	116.00	1037
SARI cases	220.12	161.49	1	172.50	1375

Figure 3.1 shows co-occurring temporal variation in pollution and notified morbidity. The series are heterogeneous in scale and temporal shape; visual co-movement alone does not distinguish pollution effects from seasonal or reporting processes.

3.2 Primary adjusted associations

ILI was the only outcome with a positive association whose 95% CI excluded the null (Table 3.2; Figure 3.2). A 10 $\mu g/m^3$ higher $PM_{2.5}$ concentration was associated with 17.3% higher ILI notification counts. The estimate for SARI was positive but imprecise. The under-five pneumonia estimate was below one but compatible with the null. The inverse TB estimate was statistically precise but biologically and methodologically ambiguous; it must not be interpreted as evidence that particulate pollution protects against TB.

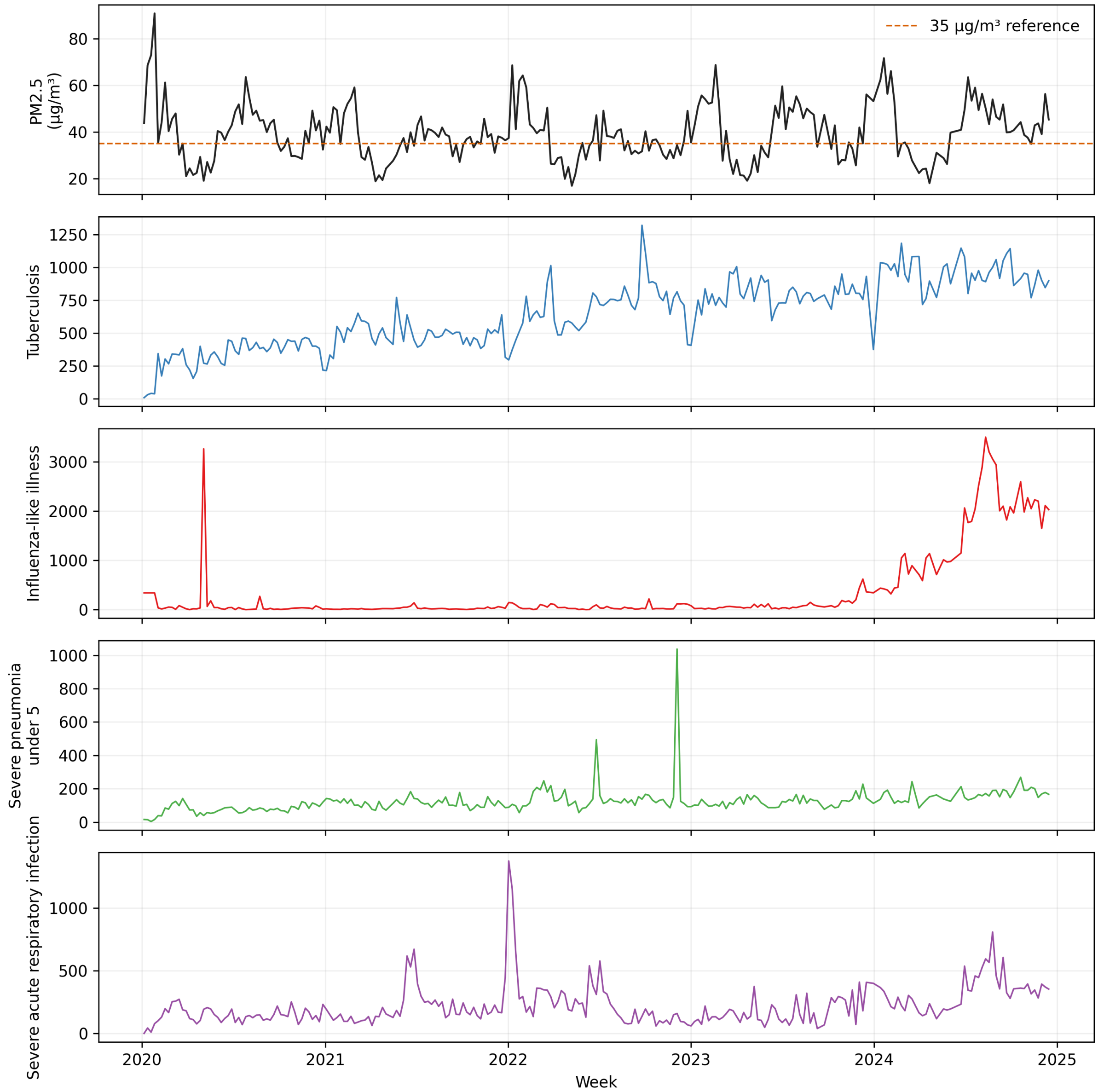

Figure 3.1: Weekly $PM_{2.5}$ concentrations and respiratory notification counts. The dashed line is the notebook's 35 $\mu g/m^3$ visual reference, not a causal threshold used in modeling.

Table 3.2: Primary adjusted Poisson-model associations per 10 $\mu g/m^3$ higher $PM_{2.5}$.

Outcome	IRR	95% CI	p value	AIC
TB	0.952	0.924–0.981	0.001	9616.2
ILI	1.173	1.030–1.335	0.016	85163.4
Severe pneumonia, under 5	0.956	0.908–1.005	0.080	7043.4
SARI	1.038	0.950–1.135	0.410	20987.4

All models used 248 weeks and adjusted for temperature, relative humidity, precipitation, calendar month, and linear time. HC3 robust standard errors were used.

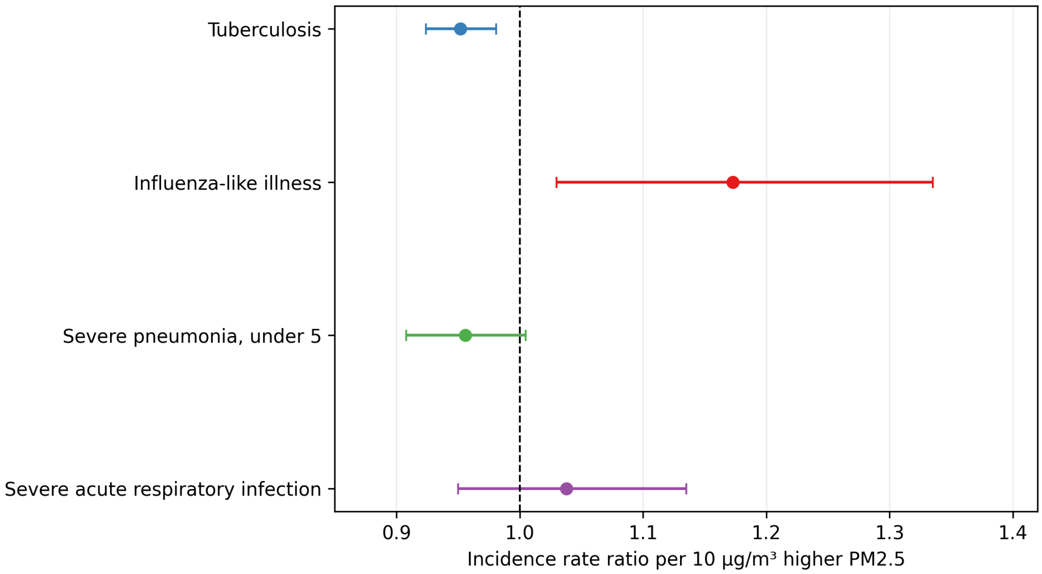

Figure 3.2: Primary adjusted incidence rate ratios. Points are IRRs per 10 μg/m^3 higher $PM_{2.5}$; horizontal bars are 95% confidence intervals.

3.3 Lagged and climate associations

The 0–4-week cumulative association was positive for ILI (IRR 1.192, 95% CI 1.032–1.377) and was compatible with the null for TB (0.977, 0.952–1.002), severe pneumonia (0.969, 0.903–1.039), and SARI (1.043, 0.951–1.144) (Figure 3.3). No single lag pattern by itself supplies strong evidence of a consistent exposure response for every outcome.

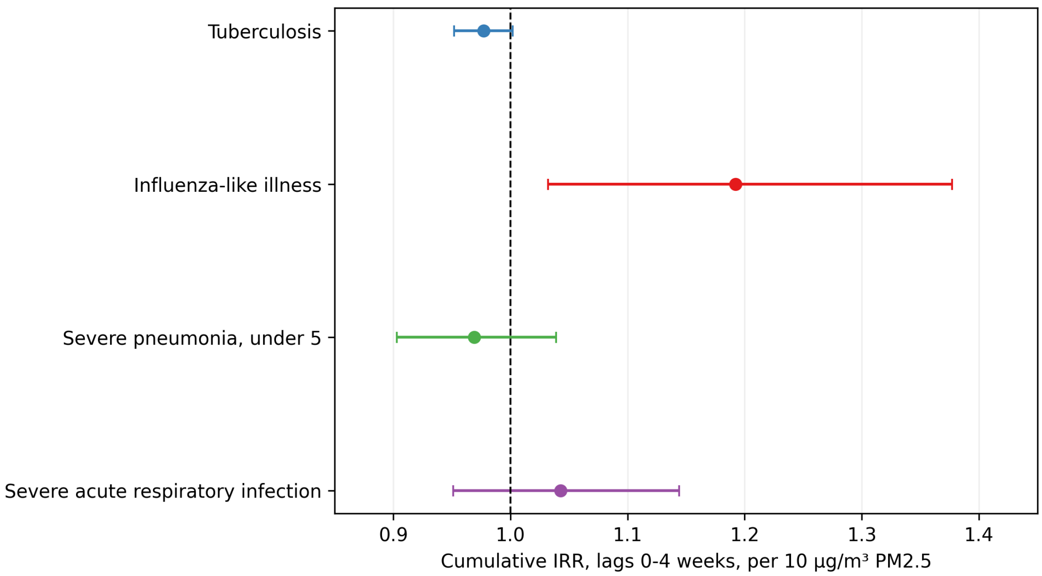

Figure 3.3: Cumulative 0–4-week adjusted lag associations per 10 μg/m^3 higher $PM_{2.5}$.

Temperature was inversely associated with TB notifications (IRR 0.945 per 1°C, 95% CI 0.909–0.983; $p = 0.005$). All other temperature and humidity estimates were imprecise at the 0.05 level. This pattern reinforces that weather and time-varying system factors are important potential confounders and not merely nuisance covariates.

3.4 Seasonal and spatial patterns

Seasonal profiles varied by outcome (Figure 3.4). In the notebook's seasonal totals, ILI and SARI notifications were highest during June–August, whereas under-five severe pneumonia had its largest total in December. Such temporal structure motivates adjustment for month but cannot eliminate residual seasonal confounding.

Division-level monitored means ranged from 32.26 μ g/m^3 in Nakawa to 54.33 μ g/m^3 in Rubaga (Figure 3.5). Monitoring density varied (one to three sensor sites by division), as did the number of health abstraction records. Therefore, geographic contrasts are an environmental and service-data description, not adjusted small-area disease-risk estimates.

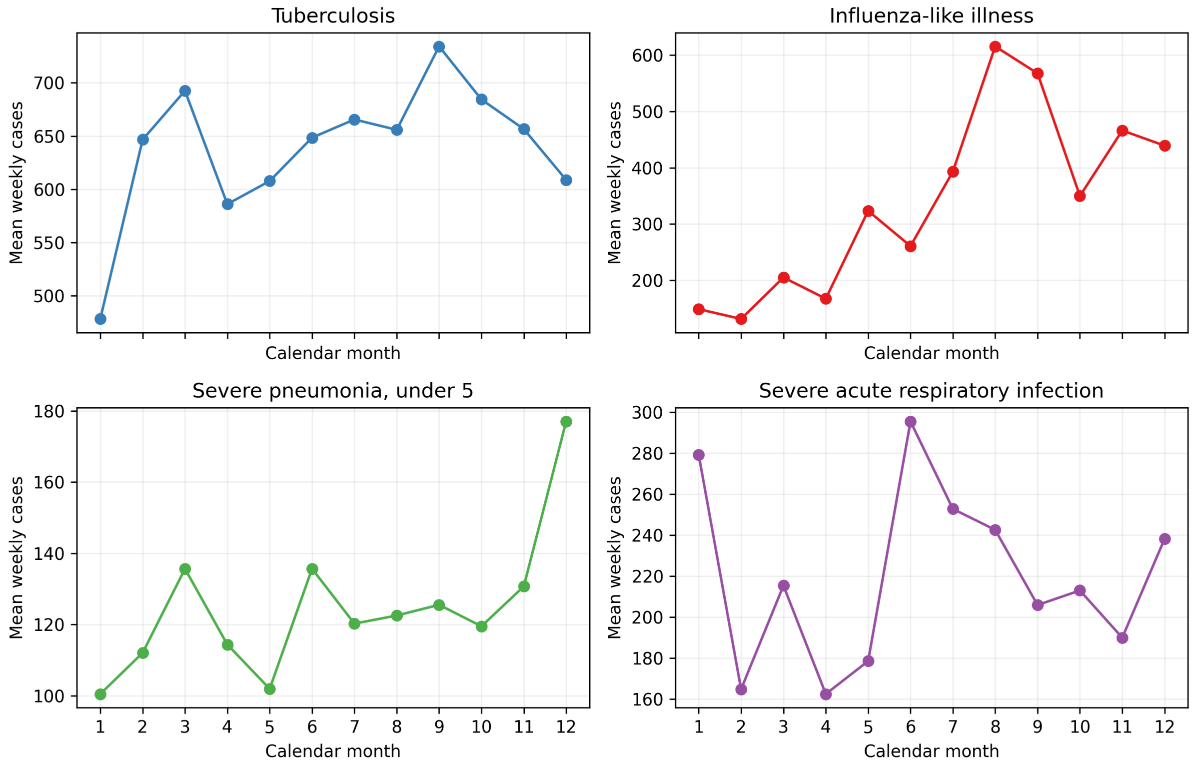

Figure 3.4: Mean weekly notifications by calendar month. Each panel has its own outcome scale.

3.5 Sensitivity analysis

The positive ILI association changed little in both sensitivity analyses (Table 3.3). The TB association attenuated toward the null with a one-week exposure lag. The remaining outcomes remained non-significant. These checks demonstrate some within-notebook stability but do not resolve structural biases in exposure measurement, diagnostics, reporting, or unmeasured confounding.

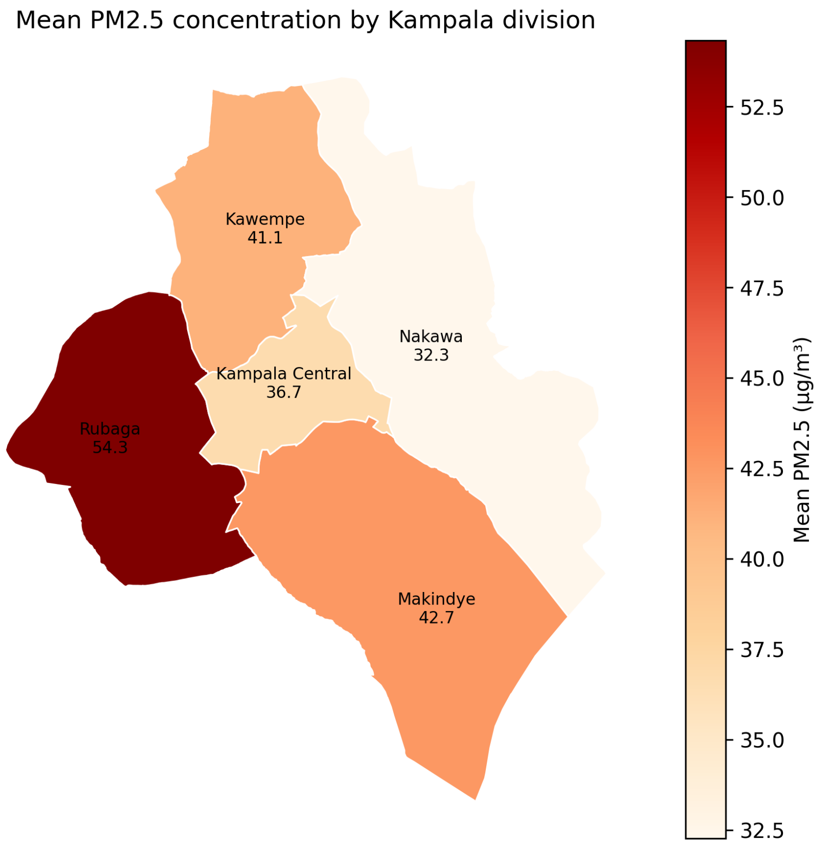

Figure 3.5: Mean monitored $PM_{2.5}$ concentration by Kampala division in the supplied spatial monitor exports.

Table 3.3: Sensitivity analyses for $PM_{2.5}$ associations, IRR per 10 μg/m^3 higher exposure.

Outcome	Specification	N	IRR	95% CI
TB	Primary	248	0.952	0.924–0.981
TB	Exclude highest 5%	235	0.967	0.941–0.994
TB	One-week lag	247	0.978	0.955–1.002
ILI	Primary	248	1.173	1.030–1.335
ILI	Exclude highest 5%	235	1.171	1.023–1.341
ILI	One-week lag	247	1.171	1.020–1.344
Pneumonia, under 5	Primary	248	0.956	0.908–1.005
Pneumonia, under 5	Exclude highest 5%	235	0.981	0.915–1.051
Pneumonia, under 5	One-week lag	247	0.961	0.904–1.022
SARI	Primary	248	1.038	0.950–1.135
SARI	Exclude highest 5%	235	1.044	0.962–1.133
SARI	One-week lag	247	1.007	0.939–1.081

Chapter 4

Discussion

4.1 Interpretation

The strongest internally consistent result was a positive association between weekly $PM_{2.5}$ and ILI notifications. Its primary and cumulative lag estimates were above one, and the result persisted after removal of high-pollution weeks and after a one-week lag. This pattern is compatible with, but does not prove, a respiratory morbidity effect of particulate pollution.

The inverse TB association requires particularly cautious interpretation. TB notifications are shaped by chronic disease processes, care pathways, diagnostic capacity, treatment programs, and reporting conventions that need not respond to weekly ambient exposure in a simple way. An apparent inverse short-term ecological association can arise from residual temporal confounding, exposure misclassification, changes in services, non-comparable reporting, model misspecification, or chance. It is not a basis for a protective interpretation.

The absence of conventional statistical significance for SARI and under-five severe pneumonia is not evidence of no effect. Confidence intervals include potentially relevant positive and negative effects, while outcome ascertainment, small effective temporal variation, aggregation, and limited covariate control constrain precision and validity.

4.2 Strengths

Strengths include a multiyear weekly series, explicit linkage of environmental and surveillance sources, multiple respiratory outcomes, robust standard errors, lag analyses, and transparent sensitivity checks. The chapter also distinguishes exploratory analyses from primary inference and retains the complete no-mortality analytic scope.

4.3 Limitations

First, this ecological analysis cannot link personal exposure to an individual's outcome. Second, citywide or division-level averaged monitor data can misclassify individual exposure. Third, missing calendar weeks, mean imputation, and reliance on routine notifications can introduce artifacts. Fourth, key confounders were unavailable, including population denominators, healthcare utilization, diagnostic testing, mobility, respiratory viral circulation, socioeconomic conditions, indoor pollution, and intervention timing. Fifth, count models did not include a population offset, so the IRRs concern notified counts. Finally, the simplified linear lag model does not characterize non-linear concentration-response relationships.

4.4 Implications and recommendations

Kampala should strengthen interoperable environmental and health surveillance, preserve facility-level denominators and diagnostic metadata, and maintain calibrated monitors with documented coverage. Future research should pre-specify causal estimands, use population offsets and finer spatial-temporal exposure assignment, evaluate overdispersion and autocorrelation, include non-linear splines and formal distributed-lag cross-bases, and validate findings against independent data. Pollution-reduction decisions should use the broader health evidence base and local feasibility assessments rather than the chapter's scenario calculations alone.

Chapter 5

Conclusion

Across 248 weekly observations, higher ambient $PM_{2.5}$ was associated with higher ILI notifications in adjusted and cumulative-lag models. Associations for SARI and under-five severe pneumonia were imprecise, while the inverse TB association is not plausibly interpretable as protection. These findings are hypothesis-generating observational evidence. They justify improved surveillance and more rigorous prospective analysis, not causal claims about individual risk or predicted benefits of a specific intervention.

Appendix A

Reproducibility and analytic specifications

The primary reproducibility artifact is `analysis_no_deaths.ipynb`. It reads project-supplied climate, $PM_{2.5}$, and routine respiratory data; removes all mortality outcomes; creates the merged weekly analytic file; and calculates the reported outputs. The companion script `generate_thesis_assets.py` regenerates the no-mortality merged input and the figures used in this chapter. Re-running the complete notebook may change stochastic neural-network outputs unless random seeds and software versions are fixed.

Appendix B

Exploratory analyses

B.1 Interrupted time-series reference

The notebook set 29 May 2022 (the series midpoint) as an intervention reference. No documented external intervention was identified in the supplied material. Consequently, any before–after or interrupted-time-series contrast is a synthetic modeling reference and is not reported as an intervention effect.

B.2 Forecasting diagnostics

The fusion LSTM model used 188 training and 48 test samples for each outcome. Its out-of-sample performance was poor: TB RMSE 364.25 and $R^2 = -6.09$; ILI RMSE 1721.38 and $R^2 = -2.71$; under-five severe pneumonia RMSE 287.34 and $R^2 = -61.21$; and SARI RMSE 310.67 and $R^2 = -3.76$. Negative R^2 values show that these configurations performed worse than a mean-baseline benchmark on the test data. They are not suitable for operational forecasting without redesign, tuning, external validation, and uncertainty evaluation.

B.3 Scenario calculations

The notebook considered 15%, 20%, 25%, and 35% hypothetical $PM_{2.5}$ reductions and calculated expected prevented cases using a fixed assumed relative risk of 1.08 per 10 $\mu g/m^3$ change. This assumption was not estimated separately for each outcome in the primary models and ignores uncertainty, feasibility, costs, exposure distributions, and causal identification. The calculations should therefore be treated as illustrative arithmetic, not policy-effect estimates.